

Proposed Solution

Date	04 October 2026
Team ID	POCKETSMART-TEAM
Project Name	PocketSmart AI – Smart Budget & Recommendation Assistant using Gemini Models
Maximum Marks	3 Marks

Proposed Solution Overview

PocketSmart AI is a web-based budget and recommendation assistant. After signing in, a user chooses a Home Interior, Party or Jewelry planner and enters a total budget with a few requirements. The FastAPI backend validates the request and sends a structured prompt to Gemini. The reply is parsed into a breakdown with estimated INR prices, a calculation table and the remaining budget, and each item is given shopping search links. If Gemini cannot respond, a rule-based estimate is shown and labelled as such.

Core Components

1. **User Interface:** Landing, registration, sign-in, dashboard, three planner pages and history page.
2. **FastAPI Backend:** Routes, authentication, session handling and validation (main.py, auth.py, models.py).
3. **Gemini Integration:** Prompt building, model fallback with retry, JSON extraction and image input (gemini_utils.py).
4. **Shopping Links:** Search URLs for Indian platforms generated from the AI's search terms.
5. **Recommendation History:** Per-user list of past plans that can be reopened in full.

User Flow

```
Register → Sign In → Dashboard → Choose Planner → Enter Budget & Details
      ↓
    Validation → Gemini (3 models tried) —fails→ Static fallback estimate
      ↓
Breakdown + Shopping Links → Saved to History → Reopen from History
```

Team Responsibilities

Madhumitha S – FastAPI and authentication; Chitabatini Tanushree – frontend and dashboard; Srija Venkataraman – Gemini integration; Praveenkanth M – validation and testing; Karthikha PR – documentation and demo preparation.